



INTERNATIONAL ORIENTEERING FEDERATION

IOF TV MANUAL

For broadcasters

[Version 1.2, 19/08/2022]

Issued by: [IOF TV Advisor, IOF Office]

1. INTRODUCTION

The purpose of this document is setting up of the standards and guidelines for Host-broadcasters and broadcasters of the major live orienteering TV and streaming productions. Every hostbroadcaster and broadcaster of the event organized by the IOF should follow the rules and instructions of this manual wherever it is possible and should inform representatives of the IOF immediately about potential problems and collisions.



2.0 BASIC PRINCIPLES

2.1 EQUALITY AND FAIRNESS

Every TV production should be fair and equal to any athlete and any nation and should truly describe progress of the competition. It is essential to focus on favourites of the race and current leaders, but it is possible and welcome to present athletes from rightsholders countries, other countries and developing countries wherever it is possible and if it doesn't corrupt proper storytelling. Presentation of the athletes from developing countries should be provided with full respect to their orienteering skills level.

2.2 TRUE DESCRIPTION

Every TV production should truly describe terrain, surface, elevation, weather and other important parameters of the race. It is also important to describe orienteering as a sport where athletes choose their own route choice from control to control. This should be done by using right moments of the GPS tracking replays/live shots, by positioning more cameras in the a row between or close to controls, by using running/following live cameras. Every hostbroadcaster should avoid such kind of presentation where orienteering looks similar to steeplechase or cross country running.

2.3 CONTROL TO CONTROL PRINCIPLE

Wherever it is possible, use of „control-to-control“ method of coverage is preferred. This method means, that the athlete is shown continually for the whole period of approach the first control, punching this control, running to next control and punching that control.



Example of camera positioning:

CAM 1 – before first control with good visibility both to control and to expected inrun zone

CAM 2 – between first and second control

CAM 3 - close to third control with a good view to the control and to the expected out-run zone

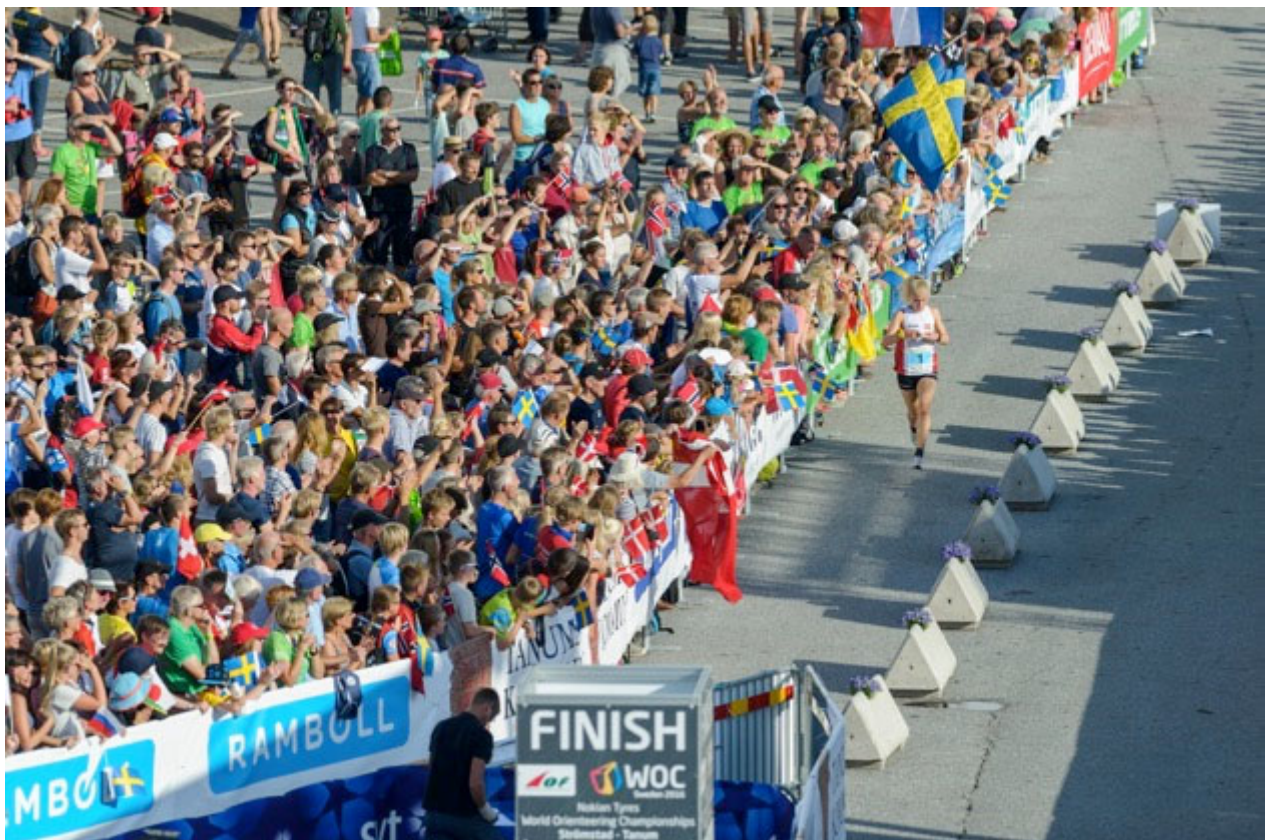
NOTE:

wherever it is applicable, cameras should be positioned in a way, that doesn't make the navigation easier for the athlete. This applies also to the camera cables layout, that shouldn't allow following from control to control. Camera operators, photographers and other personnel present in the competition areas should be dressed and behave in a manner, that doesn't affect competition fairness.

2.4 ATTRACTIVITY

Every host broadcaster should describe orienteering as an attractive, modern and dynamic sport. In order to fulfil this task it is important to:

- show orienteering in fast and attractive parts of the course
- avoid presentation of non good looking places such as suburbs, dirty corners, etc.
- keep an eye on arena look and feel
- Present commercial elements in a proper way with a good visibility but in a biased manner
- Include as many fans and spectators as possible



3.0 INTERVAL RACES

PRODUCTION PHILOSOPHY

3.1 BASIC DESCRIPTION

The basic production idea of the interval races coverage is based on the race and course simplification and repetition. There are following mandatory areas of the TV coverage:

- start
- split 1
- split 2
- finish

In order to increase attractivity and dynamics of the production there might be also another areas covered, especially (but not only):

- quarantine
- warm up
- coaching zone
- arena passage
- additional split
- leader board (chair)
- team zone
- cool down area
- prize giving area

Split 1 should be positioned cca after one third of the expected race time

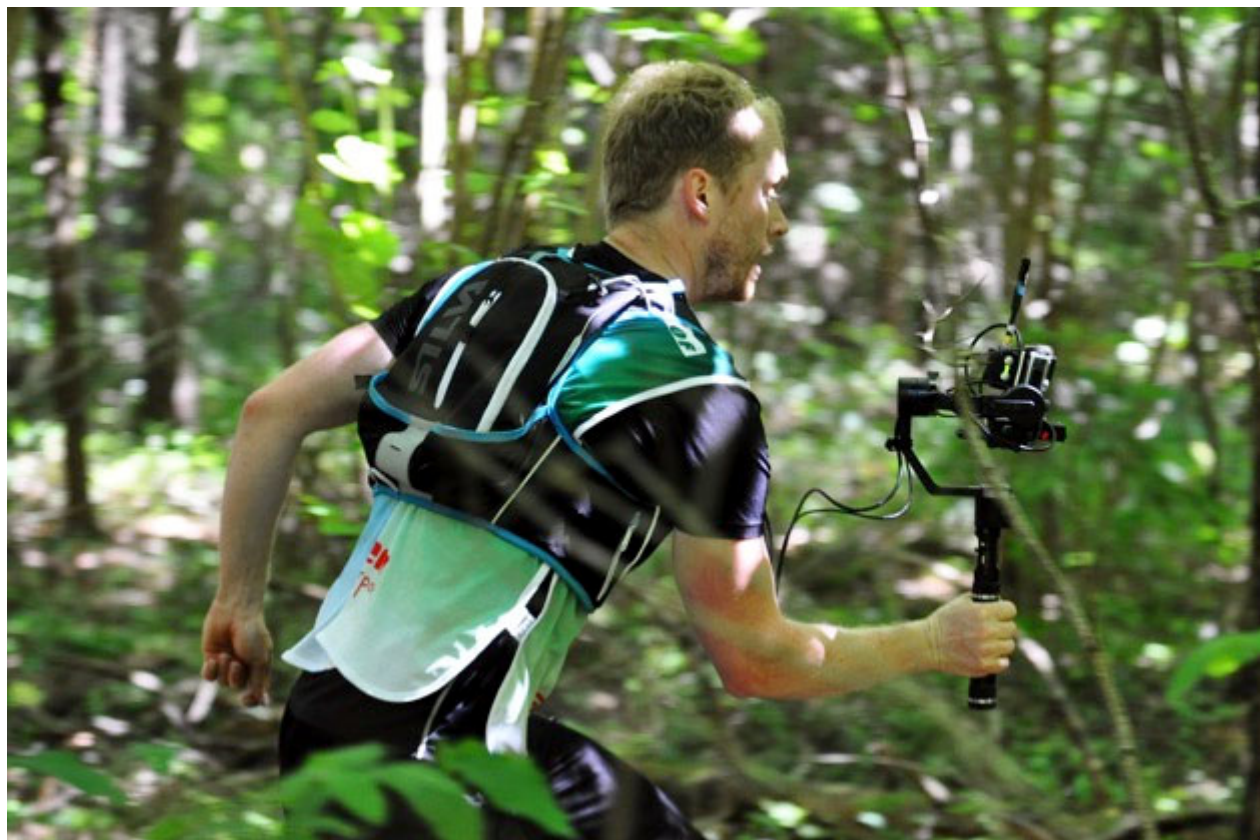
Split 2 should be positioned cca after two thirds of the expected race time

Inrun to the finish should be captured by sufficient number of cameras .

3.2 PRODUCTION SETUP LEVEL A

Production setup level A should be mandatory for all World Orienteering Championships.

CAM 1	warm up zone
CAM 2	start close up
CAM 3	start outrun
CAM 4	split 1 first control inrun and punch
CAM 5	split 1 transition
CAM 6	split 1 second control punch and outrun
CAM 7	split 2 first control inrun and punch
CAM 8	split 2 transition
CAM 9	split 2 second control punch and outrun
CAM 10	finish inrun to the last control
CAM 11	finish near last control
CAM 12	finish transition to finish line
CAM 13	finish wide shot
CAM 14	finish handheld



Cameras at start, finish arena and at least 1 split time shall be covered with own infrastructure (cables) to secure broadcast in case of failure of external networks.

At least 2 cameras should be „runcams“ – stabilized minicameras with RF video transmitter (i.e. GoPro 7 + HeroCast or Sony RX + IMT Microlight 2)

If there are additional cameras available, they can be positioned as described in paragraph 3.1. In addition to mentioned above, host broadcaster may use i.e.

- another camera before the finish
- closeup finish camera
- extra cameras for split times coverage

Any additional grip or special camera is welcome add-on. This may be i.e. jimmyjib, railcam, trackcam, technocrane, drone, SSM or USM cameras, RF cameras, beautyspot elevated cameras, comcams, etc.



3.3 PRODUCTION SETUP LEVEL B

Every IOF organized event should use at least Production plan B standards to ensure sufficient broadcast quality. Standard B is a cost effective solution using less production resources.

- CAM 1 start
- CAM 2 split 1 first control inrun and punch
- CAM 3 split 1 transition
- CAM 4 split 1 second control punch and outrun
- CAM 5 split 2 first control punch and transition
- CAM 6 split 2 second control inrun and punch
- CAM 7 finish near last control

CAM 8 finish transition to finish line

CAM 9 finish wide shot

CAM 10 finish handheld

At least 1 camera should be „runcam“ – stabilized minicamera with RF video transmitter (i.e. Gopro 7 + HeroCast or Sony RX + IMT Microlight 2)

Cameras at start, finish arena and at least 1 split time shall be covered with own infrastructure (cables) to secure broadcast in case of failure of external networks.

4.0 MASS START RACES

PRODUCTION PHILOSOPHY

4.1 BASIC DESCRIPTION

As there is highly anticipated that mass start (and pursuit) races will include forkings and butterflies (= areas, which camera coverage is confusing), the basic production philosophy is based on two elementary premises:

- show common legs and controls
- show non-common controls and legs in the GPS tracking
- runcam may be used also at forked legs

Note:

camera positioning and numbering is indicative. It is essential, that every host broadcaster will make his own choice based on experience, actual race setup and IOF guidelines.



4.2 PRODUCTION SETUP LEVEL A

Production setup level A should be mandatory for all World Orienteering Championships.

- CAM 1 start HH
- CAM 2 start wide shot
- CAM 3 start outrun
- CAM 4 common controls out of arena
- CAM 5 common controls out of arena
- CAM 6 common controls out of arena
- CAM 7 common controls out of arena

CAM 8 common controls out of arena

CAM 9 finish near last control

CAM 10 finish transition to finish line

CAM 11 finish wide shot

CAM 12 finish handheld

RUN 1 Runcam A

RUN 2 Runcam B

Both Runcams should be positioned in a way, that allows its use more than 1x per leg – wherever it is possible.

4.3 PRODUCTION SETUP LEVEL B

Every IOF organized event should use at least Production plan B standards to ensure sufficient broadcast quality. Standard B is a cost effective solution using less production resources.

CAM 1 start HH

CAM 2 start wide shot/outrun

CAM 3 common controls out of arena

CAM 4 common controls out of arena

CAM 5 common controls out of arena

CAM 6 common controls out of arena

CAM 7 finish near last control

CAM 8 finish transition to finish line

CAM 9 finish wide shot

CAM 10 finish handheld

RUN 1 Runcam

5.0 CORPORATE IDENTITY

In order to strengthen brand identity and to unify TV production look and feel IOF created set of CI visual and audio elements. Every broadcaster should use those elements in a proper way described below. IOF will deliver download link to every host broadcaster separately at least 1 month before the Event.

5.1 OPENERS, JINGLES

Official IOF opener should be used always at the beginning and at the end of the live TV production or streaming. If the event has its own opening sequence, the Official IOF opener should be played before (at the beginning of transmission) or after (at the end of transmission).

5.2 WIPES

Official IOF wipe should be used for every live/replay transition as well as other transitions (live/picture in picture, live/GPS tracking, etc...)

5.3 GRAPHICS

Every broadcaster is obliged to use official IOF live TV graphics set.

5.4 MUSIC

At the moment there are no special rules for use of music.

6.0 GPS TRACKING

GPS tracking is an elementary tool for any live TV orienteering production. The system is based on GPS tracking unit, that continuously tracks the exact position of the runner using GPS satellite system and sends online those data to the server using GPRS or 3G/4G network. Data from server are visualized during the race in the ways described in paragraph 6.2.

6.1 GENERAL GPS TRACKING USER SETUP

There should be one (production setup B) or two (production setup A) GPS tracking hardware systems connected via HD SDI breakout box. Default resolution of the GPS tracking video output should be HDTV 1920x1050 at 50fps (or at 25i).

6.2 GPS TRACKING GRAPHICS

This section should describe preferred and mandatory graphics for GPS-tracking:

- Line width: 12 pixels
- Tail length: According to starting interval and to the extent possible consistent throughout the race
- Coloring of letters: Identical to dot and tail. To the extent possible consistent throughout the race
- How to show forking: Identify control forking alternative by letter on map, and for each runner display letter corresponding to his forking alternative.
- Use of flags: In leaderboard, and in relay competitions next to tracking dot.

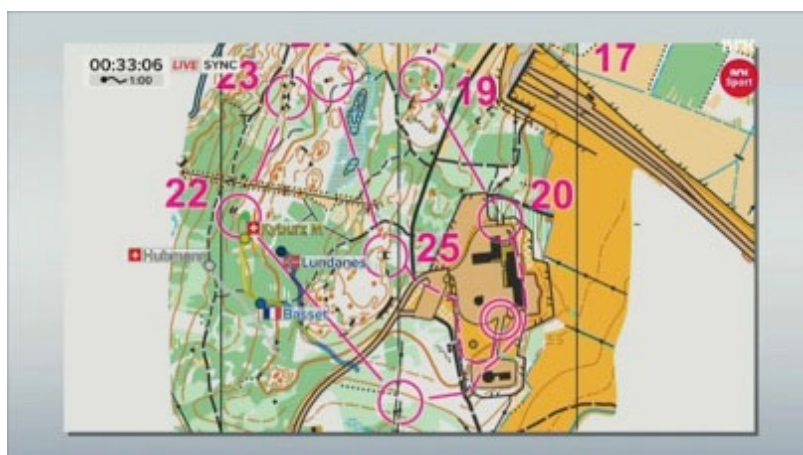
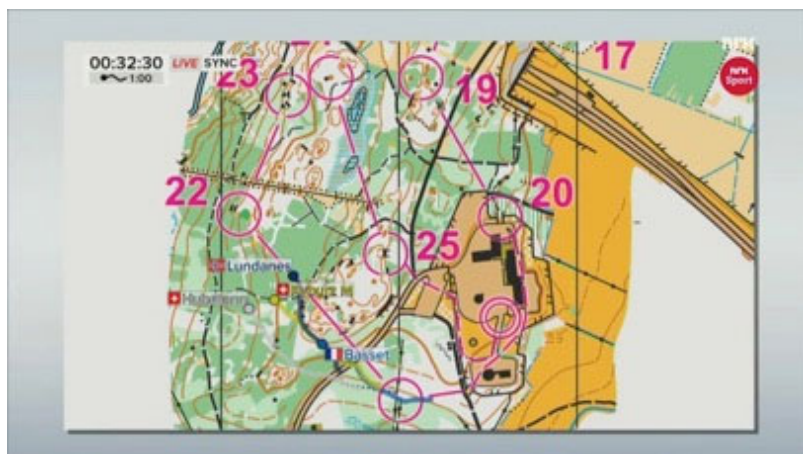
6.3 BASIC GPS TRACKING USE WORKFLOWS

Live tracking

Selected units are displayed in a real time without any offset/sync. This workflow is used mostly in mass start races, relays and pursuits

Live sync tracking

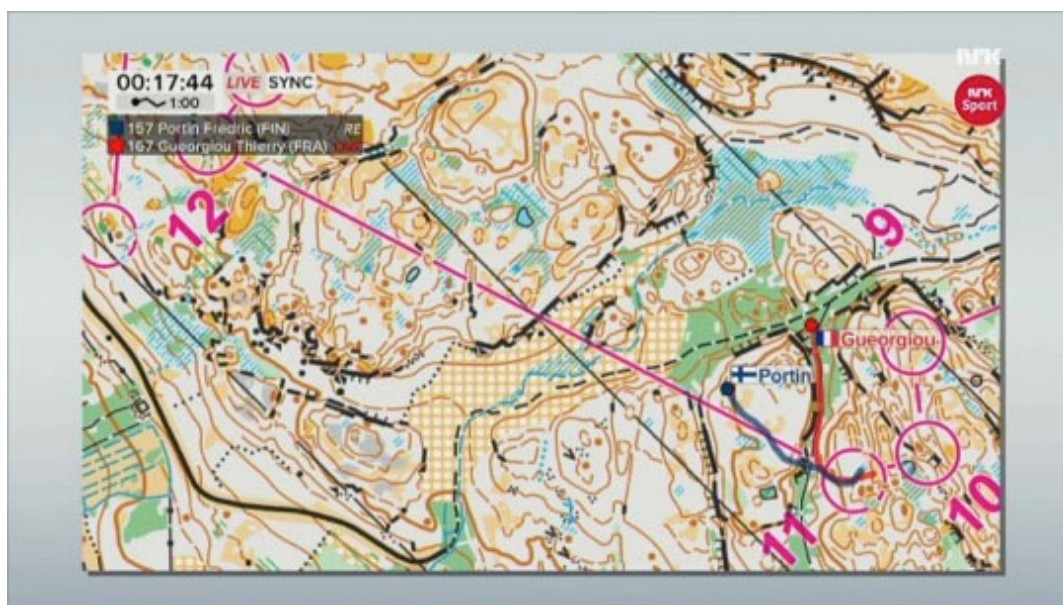
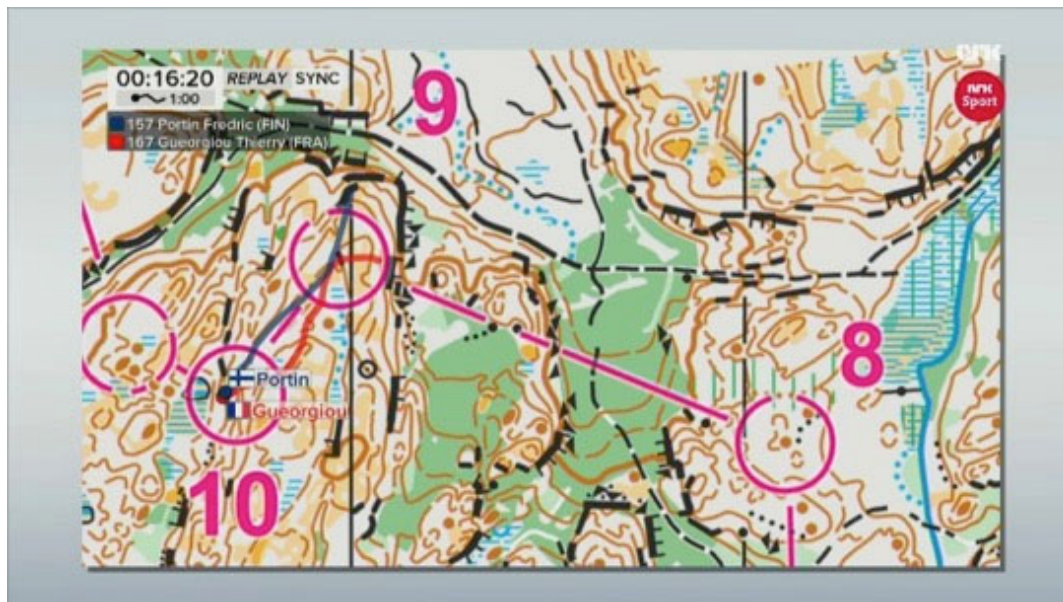
One unit (last starting competitor) is displayed in a real time. Another unit (competitors starting before real time displayed unit) are offset to the same start time and point as the real time displayed unit. Usually only two units are shown, but in the final stages of races it can make sense to show more units. This workflow is used i.e. at the last stages of interval start races or to replay mistakes etc.



Example snapshots from live sync between four runners in the final stage of WOC 2016 Middle Men.

Replay to live

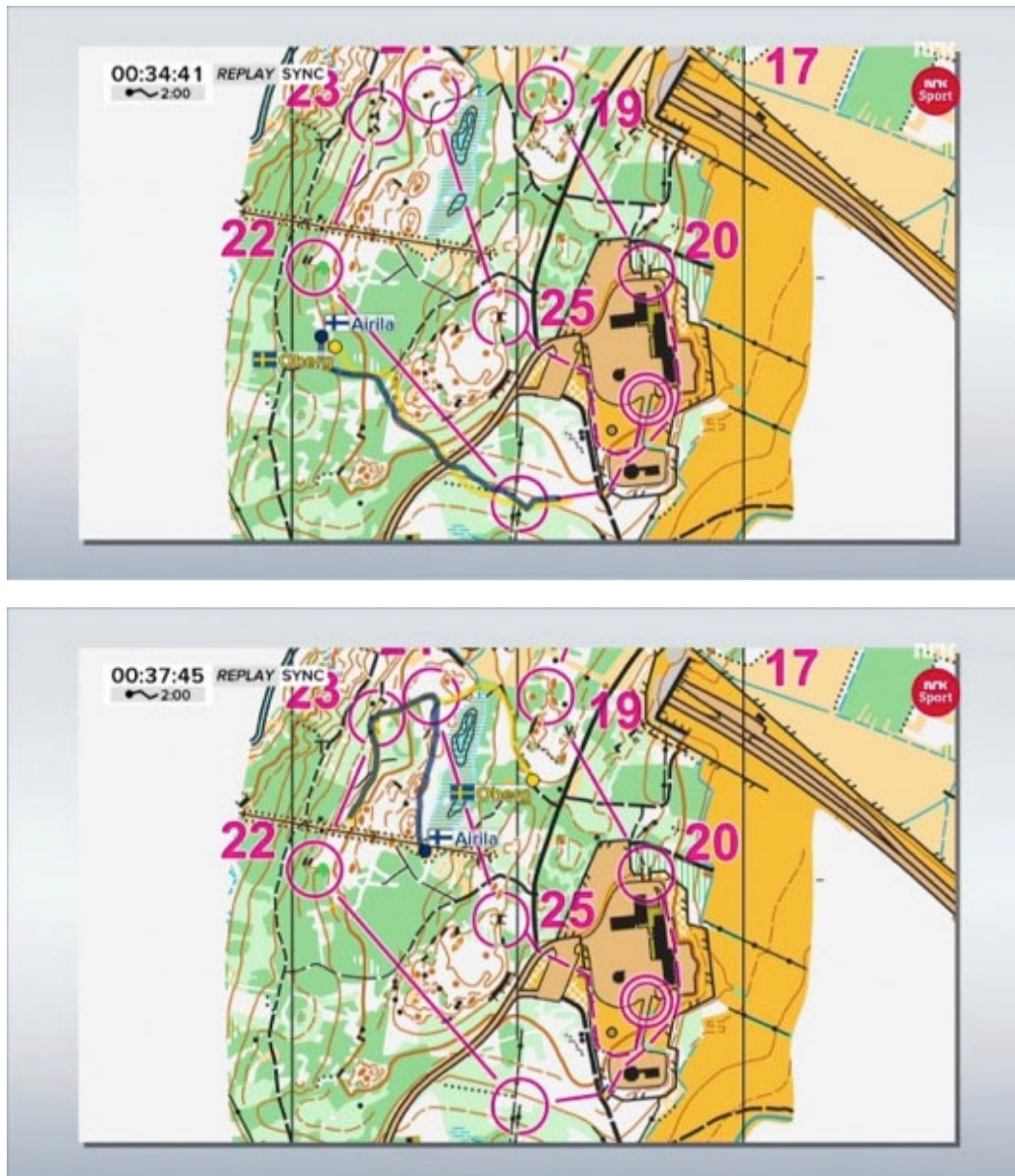
One or more units are displayed from the moment several seconds or minutes before live in replay mode. Once unit(s) reach end of replay mode, it turns to live mode seamlessly. Used both for individual and mass start races.



Example snapshots from replay-to-live between two runners from WOC 2016 Middle Men.

Sync replay

Two or more units are displayed in replay mode using the same start time and position offset. May be used for showing mistakes, decisive or interesting route choices both in interval and mass start races.



Example snapshots from sync replay between two runners from WOC 2016 Middle Men.

Current runner against current leader

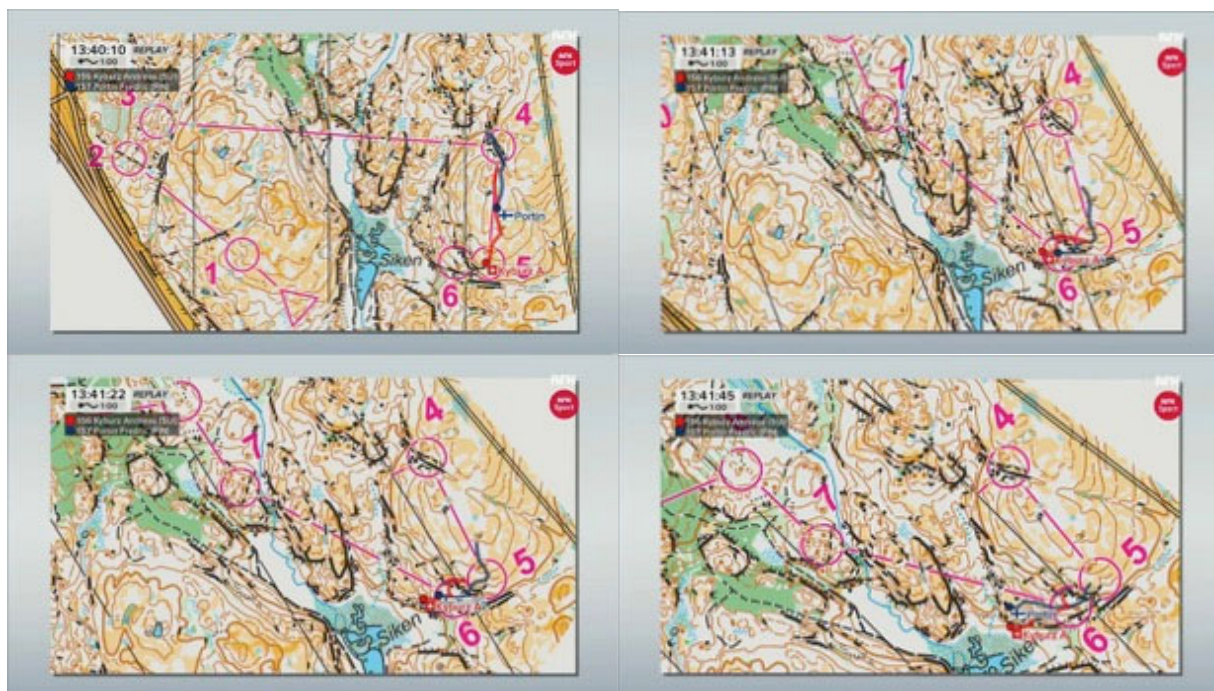
In this case there are two units compared – current runner (i.e. after the TV leg) and current leader. Sync replay workflow is used for this case

Single replay

Only one unit is displayed in replay mode. This workflow is used for explicit display of any specific cases, i.e. decisive mistake. Can be used in any format of the race. Note however that it may often be better to show decisive mistakes as SYNC against leader in order to show a reference for the decisive mistake.

Effects to improve viewer experience of GPS replay

It may be difficult for the viewer to understand the GPS-tracking if the course is complex with many crossings. For these cases it is preferable to show only parts of the course on the map and blend in other parts when necessary. Also, rotation of the map to specific legs during replay may give the viewer a better understanding of the tracking. See example in the below illustration.



Example snapshots from GPS tracking from WOC 2016 Middle Men where the first six controls are shown during the first part of the GPS-replay, whereas controls 1 to 3 are hidden and controls 7 to 9 are shown when the first runner arrives at control 5. Note also the slow rotation of the map when the first runner arrives at control 6.

6.4 WAYS OF GPS TRACKING IMPLEMENTATION

Every producer should keep in mind, that GPS tracking is an important storytelling and educational tool and should use and implement it into the storyline in a proper manner, easy to understand for everybody. The following basic rules should be followed:

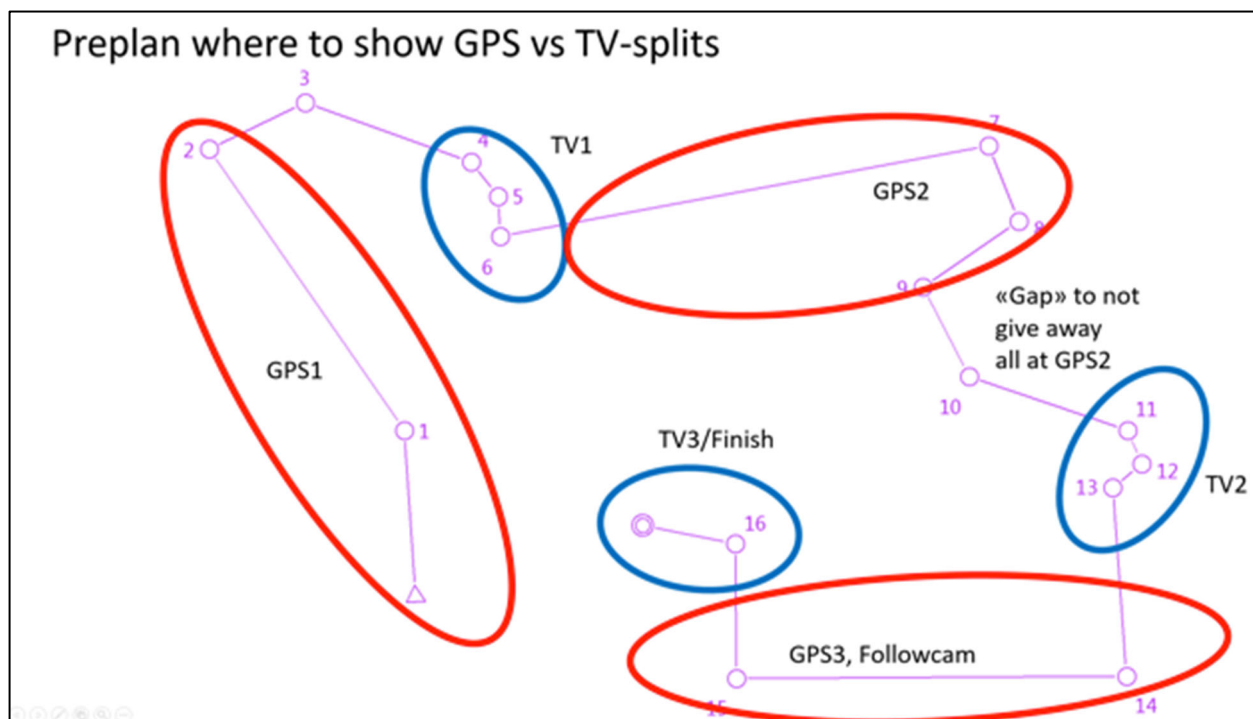
- use proper speed of animation (not too fast)
- choice of GPS tracking should be logical and consistent together with other storytelling elements (i.e. show athlete live on camera, then include his/her GPS tracking)
- for complicated replays, allow commentators and viewers to adapt and understand the case
- the number of athletes shown should always be reduced to a minimum for a clear overview.

RELAY, SPRINT RELAY, MASS START, PURSUIT

Essential mode for these competition formats is „live tracking“, it is preferred to use „replay to live“ after longer passages of live TV coverage (i.e. relay changeover). There must be clear indication of destination at forked parts of the course implemented on GPS tracking (flags, letters, etc). Mistakes and interesting route choices/comparisons may be shown in „replay“ mode.

INTERVAL START RACES

There should be predefined expected interesting parts of the courses dedicated for GPS tracking presentation. These parts should be primarily presented in „Current runner against current leader“ or in „sync replay“ mode. At the final stages of the race „live sync“ may be used. Mistakes and interesting route choices/comparisons may be shown in „replay“ mode.



Example of how expected interesting parts of the course dedicated for GPS-tracking presentation can be planned.

7.0 SPECIAL WORKFLOWS

This chapter describes various special workflows used during live orienteering production and will be updated accordingly.

7.1 GPS/LIVE SHOTS MULTIPICTURE

In this mode live shots from camera are presented together with live GPS tracking of the same competitor. Live camera signal should be delayed accordingly to a delay of the GPS system. It is recommended to prepare on section in the race where GPS/LIVE multipicture will be used. Live split timing shouldn't be part of this section as there are various delays and live timing shouldn't be synced. Recommended delay of GPS tracking system: max 10 seconds

8.0 RUNNING ORDERS

Every running order (show build-up) should include following elements:

IOF official opener/stinger

20-30 sec. IOF opener should be used in cases, where Event opening sequence doesn't contain IOF corporate identity elements

Event opening animation

15-20 sec. Event opening animation should be used for every single live production of orienteering.

Postcard

60 sec. Postcard should describe in attractive and readable manner location of the event, its monuments and general feeling.

Basic event info

60 sec, In this section information about the race (length, nr. Of controls, est. winning time, number of participants), weather and other valuable information should be provided in as a TV graphics.

Course map animation

2 min. The whole course animation including forking (where applicable) should be presented before every single race in a proper (= not too fast) manner that allows commentators and viewers understand the course. Hill shading is preferred. TV splits should be highlighted during course map animation.

Route choices analysis

2 min. Decisive or interesting route choices analysis (1-3 per race) should be presented prior to each race. This may be combined with course video preview.

Course video preview

2-3 min. Course presentation should describe general parts of the terrain and its environment. Gopro (or other minicameras), camera on a gimbal or drone can be used for filming of course presentation. There should be always controls installed and there should be visible runner in unbiased, jersey with good contrast between jersey and terrain, i.e. official event dress code. Multipicture composition video + course with route animation cut-out is preferred solution.



Example snapshot from course video with corresponding route animation from WOC 2016 Middle Men. Note that jersey is not optimal with respect to contrast to terrain.

Start list

Start list should be presented in a speed, that comfortably allows to commentators and viewers to read and understand race participants numbers, nationality and other details.

9.0 SIGNAL DISTRIBUTION

9.1 DISTRIBUTION TO RIGHTSHOLDERS

The IOF Office handles agreements regarding distribution of video signal to the rightsholders.

The delivery point of international feed from the host broadcaster is HD 1080i50 video signal with agreed audio channels mapping over HD-SDI if nothing else is agreed.

9.2 STREAMING

Streaming signal is usually sent to IOF Vimeo platform.

Video format 1080p25

Bandwidth 4,5Mbps

Stereo audio

RMTPS protocol will be delivered by IOF representative at latest one week prior to an event.

There may be different start times for live streaming and for satellite TV distribution. In such case TV producer and streaming responsible IOF representative have to liaise suitable solution for all parties.

Every host broadcaster must have a safe streaming solution with backup. Examples of possible streaming backup configuration:

SAT –TO – STREAM

This solution is applicable for events using fully redundant DSNG distribution. Streaming provider can downlink DSNG distributed feed and create and send stream to the IOF livecenter from their HQ. Also in this case, there should be backup HW and internet line available.

DIRECT STREAMING FROM THE VENUE

Host broadcaster can stream directly from the venue to Vimeo platform if internet connectivity allows this. For this case it is mandatory to have 2 hardware stream encoders (1x main + 1x backup) running simultaneously and 2 different internet sources (see 10.2) , i.e.:

- two different fiber connections
- fiber connection and LTE modem
- LTE modem and satellite internet

The upload capacity of the main streaming feed should be at least 20 Mbps symmetric and the upload capacity of the backup stream feed should be at least 8Mbps symmetric.

Any streaming solution must be tested at least 1 week before the first live streaming of the event on both transmitting and receiving side.

IOF will provide any necessary information (streaming configuration, production setup and elements, streaming timings, etc) at least 2 weeks prior the first live streaming of the Event.

10.0 LOGISTICS AND INFRASTRUCTURE NEEDS

10.1 POWER

There should be separated power source for the TV operation. Either ground power with separate fuse/breaker or the power generator dedicated for the TV compound. Power to a commentary positions should be either taken from the same source as for the TV compound or there should be backup /UPS for the commentary positions.

Typically there are used following sockets for OB vans:

63A/3 phase 400 V/5 pin CEE socket

32A/3 phase 400 V/5 pin CEE socket

16A/3 phase 400 V/5 pin CEE socket

Estimated power consumption cca 50 kVA.

It is recommended to have electrician onsite on TV trucks setup day to be helpful to solve potential problems with connection.

Backup power source (i.e. twin power generator) increases safety of the TV operation.

10.2 INTERNET

Every organiser is fully responsible to provide stable and secure internet connection with sufficient capacity for streaming (if provided) and other TV related tasks.

Ideal network requirements for the whole event are to have 2 separate 1000/1000 mbit connections where one is dedicated to TV, GPS and timing and the other one is for public and non-live-critical event operations (like media centre). The second one is also a backup for TV.

If this is not possible to provide at the venue, compromises may have to be done, such as limited availability of public wifi etc.

10.3 SECURITY

Every organiser is responsible to provide all necessary security solutions for hostbroadcaster operations at organiser's costs. Those solutions are mainly:

- fenced TV compound for hostbroadcaster vans and trucks
- overnight security where required by host broadcaster

Exact security requirements of the hostbroadcaster together with size and location of the TV compound should be discussed and agreed between organiser and hostbroadcaster during site visits.

10.4 ACCESSIBILITY

TV working area has to be accessible for heavy trucks and other TV working vehicles and personnel. This includes not only TV compounds but also TV areas in the field of play. Critical points might be:

- bridges with insufficient capacity for heavy trucks (i.e. 20 ton)
- sharp bends and curves (for 16m long TV trucks)
- significant vertical drops in terrain access roads
- wet and soft terrain surface on access roads and TV compound
- lack of access roads (good enough at least for ATV) between TV compound and TV zones in the field of play

10.5 LOC COOPERATION

There should be a TV Coordinator from the organizer's team designated for contact with TV hostbroadcaster . This person should be responsible for any liaison between organiser and TV, should speak English, have elementary experience with TV matters, be good organizer, be accessible by phone and e-mail before and during event.

If specified in the contract, organiser should provide sufficient number of volunteers, ATVs or other means of support to hostbroadcaster.

Hostbroadcaster should inform in advance organiser about his needs and should provide organiser with all necessary documents, i.e. camera plans, cabling plans, time schedules, rooming lists, lists of support requirements, lists of scaffoldings and platforms, etc.

10.6 TIMING AND DATA

INTERMEDIATE AND PREWARNING SETUP

There should be sufficient number of TV live split times and live prewarning points.

Minimum live timing points recommendation:

Sprint: 2-3 (1-2 split times, finish)

Middle: 3 (2 split times, finish). Additional split before finish may be added

Long: 3-4 (2-3 split times, finish)

Relay: min. 3 per leg (2 in the forest, 1 in arena)

Sprint relay: min 2 per leg (1 in the competition area, 1 in arena)

There must be prewarning live timing point installed before every TV leg (usually 1 or 2 controls before the first control covered by TV cameras).

Timing and data provider is obliged to provide commentary infosystem (CIS) to the OB van (s), to the arena speaker and to every commentary position at his or organizer's costs.

LIVE TV TIMING CONNECTION

All live timing split timing points must be connected with the core of timing system either by wire or direct radio link (= constant low latency transfer). GPRS connection is not allowed for live timing split times and may be used only as a backup.

Prewarning live timing should be connected with the core of timing system the same way as live timing points (wire or direct link). Where it is not possible, GPRS connection may be used, if there is good GPRS coverage and estimated running time between prewarning and live timing point is longer than 30 seconds.

10.7 LED SCREENS

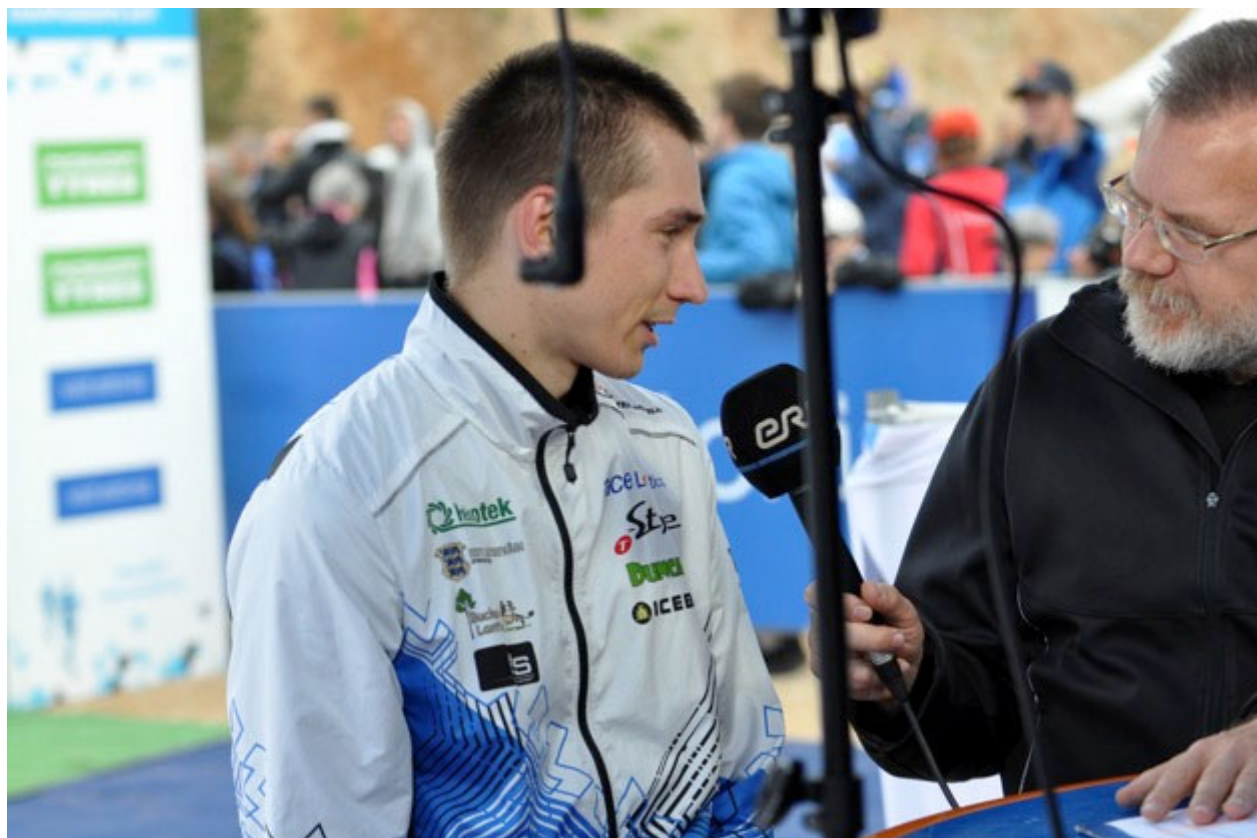
Connection between LED screens in the arena and OB van should be always provided in a manner, that allows to exclude GPS tracking signals or any other source used for maps and courses presentation (different AUX source, different M/E, signal substitution or similar).

There should be two different power sources (or fuses/breakers when use of ground power) for LED screens and TV compound in order to avoid grid overload.

Whenever it is possible, organizer should place LED screens in a way, that allows good view to spectators and doesn't give chance to athletes gain advantage by watching maps or GPS tracking (i.e. during relay exchange etc.)

10.8.MIXED ZONES

There should be mixed zone established in the finish area of each venue. Mixed zone should be well accessible both for athletes and for press representatives. Organizer will provide backdrop of sufficient size. There should be dedicated place for major broadcasters, other electronic media and written press. Space between athletes and media should be fenced.



10.9. FIELD OF PLAY

Look and feel of the field of play (competition area visible in TV and covered by still photographers) is very important aspect of success of every major IOF event. In order to reach the best possible result, these workflows are highly recommended:

- use V-Boards instead of plastic tape
- use wooden firm constructions instead of metallic fences as base for banners installation
- use firm plastic banners (i.e. makrolon) instead of canvas banners
- be sure, outrun finish zone (space after the finish line) is big enough to accommodate all photographers and TV personnel and allow finishing athlete safe and comfortable outrun and cool-down. Minimal length of outrun zone (clear space only for athletes) is 15 metres.

10.10 TOILETS

There should be toilets in every TV compound dedicated only for Hostbroadcaster or rightsholders' personnel.

11.0 IMPORTANT DEADLINES

1 year before the event

initial meeting with organizers, course setters and other involved parties. First site visit. Confirmation of arenas, TV compounds, accessibility for TV trucks, potential cable routes, terrain check, accommodation possibilities.

2 months before event

final site visit, courses and camera positions confirmation, exact logistics plan

12.0 RIGHTHOLDERS SERVICES

12.1 COMMENTATORS SERVICES

IOF will provide commentators and other representatives of accredited media with a set of documents related to the event. These documents should include (but not only):

- event press kit (basic event info, region info, terrain info, important dates)
- course setters word
- maps and routechoices description
- red group athletes bios
- camera plans
- Running orders and other hostbroadcaster's information
- Start lists
- Detailed information about are described in WOC Media Guidelines

12.2 MEDIA ASSETS LIBRARY

IOF is the owner of multimedia assets library, that is accessible to both right- and non- right holders- The library contains a number of educational clips, portraits, reviews, promos and similar resources. Use of the MMA content will be charged according to rate card.

12.3 COMMENTARY POSITIONS

Hostbroadcaster together with organizer will provide onsite commentary position if requested. Price for this service will be listed in the Event Rate Card.

Organizer will provide

- a position close to the finish on the inrun with a good view over the surroundings
- a tent with walls on three sides (or container with window facing field of play)
- table (min 2m wide) and two chairs
- 230 V Power sockets (min 4 pcs)
- internet connection
- commentary info system (CIS)

Hostbroadcaster will provide

- monitor with the Worldfeed
- commentator's box with 2 headsets (for fully equipped COP only)

11.3 RATE CARD

IOF will release Event Rate card for every Event at least 6 months prior Event start. Rate card should include prices for

- World Feed TV rights
- 90 minutes overall highlight
- 26 minutes daily highlight
- national dumps
- pre- and post- unilateral services
- fully- and partially- equipped commentary positions
- onsite TV studios and presenter's positions
- technical fees

- other services

13.0 ACCREDITATION

13.1 HOST BROADCASTER

Host broadcaster's accreditation pass should allow to a holder to access all areas and zones necessary for fulfilling his working tasks. Those areas are (but not only):

- field of play
- TV compound
- Mixed zone
- Press centre
- Award podium



Every host broadcaster representative entering field of play, team zone or similar areas should wear official IOF bib during a competition.

It is highly recommended, that host broadcaster is equipped with 2-3 „all access“ upgrade cards for cases of urgent need of unlimited access.

13.2 ENG/SOCIAL MEDIA CREWS

Host broadcaster will define zones of no access for other accredited media and will also together with photo coordinator in the media team and IOF Event Advisor of the event define dedicated zones for photo, ENG and social media teams. Zones of no access may be:

- TV legs in the forest or in urban zones
- Leader board
- Closest start area zone

Dedicated Photo, ENG and social media zones may be i.e.

- mixed zone
- finish chute/arena
- finish area
- flower ceremony
- dedicated controls near arena

General purpose of this regulation is creation of sufficient and effective workspace for all media representatives with full respect of each other. This regulation should also avoid disturbances in between hostbroadcaster and other media operations.





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